

K-01-CX-CKN-smallness-propagation-kill-test v1

ZFL-CX

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1 Mission

$$\boxed{C_F^{conc} + C_{CKN} \stackrel{?}{\Rightarrow} \int_0^{T^*} W_+ < \infty} \quad \boxed{W_+ \leq G(r, \varepsilon_0), \int_0^{T^*} G < \infty} \text{ must be proved, not assumed } r^{-3}.$$

2 Barrier

CKN: $r^{-2} \iint_{Q_r} (|u|^3 + |p|^{3/2}) \leq \varepsilon_{CKN}$ is SMALL/UPPER. F-01: $\|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$ is LOWER. No conversion. Pressure not controlled. Even if $L^\infty(Q_{r/2}) \leq Cr^{-1}$ held away, $W_+ \lesssim r_n^{-3}$, and $r_n \downarrow 0$ gives $\int r_n^{-3} = +\infty$ without rate $r_n(t) \gtrsim (T^* - t)^\beta$ which needs Type I.

3 Kill

KILLED: F-01 lower vs CKN upper, local regularity not giving integrable W_+ , r^{-3} only dimensional, still $\int W_+ = +\infty$ compatible.